

0.5-11 GHz Analog Fiber-Optic Links for the Allen Telescope Array

Preliminary Design Review—Analog

November 10, 2000

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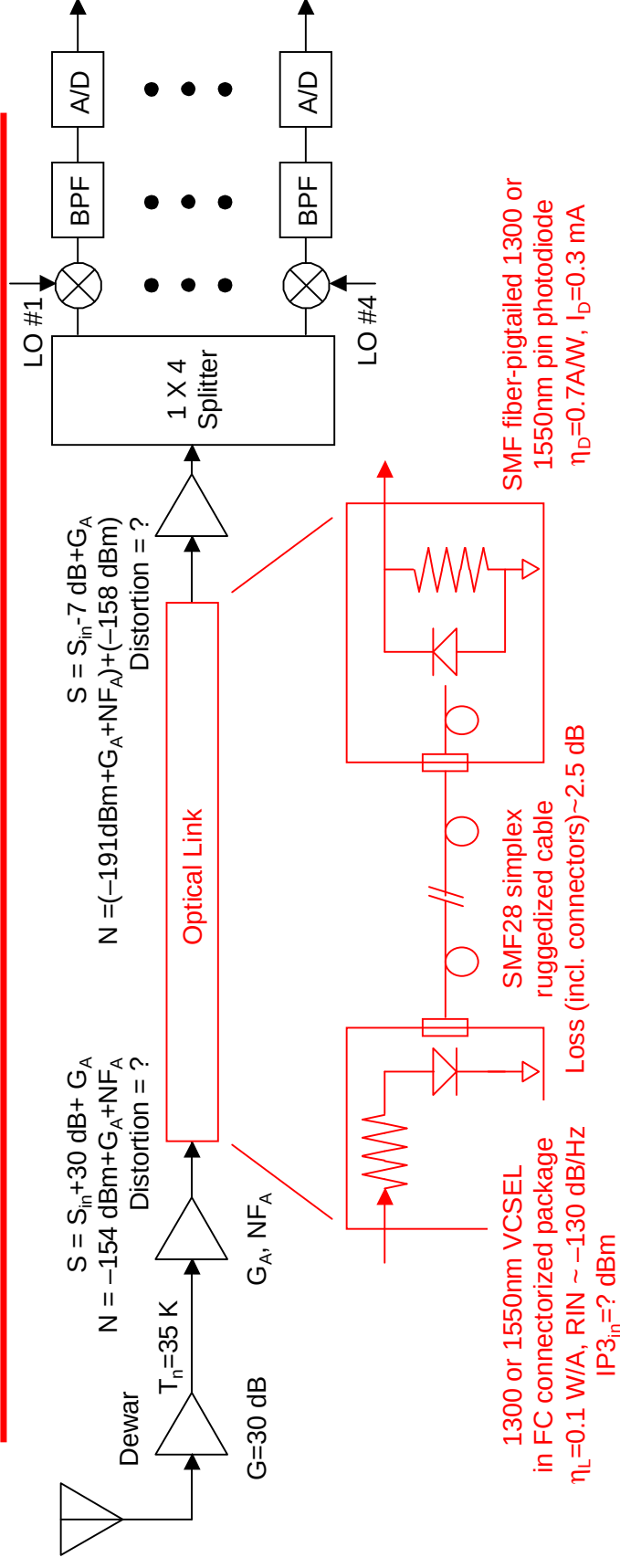
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- **Baseline approach**
 - 0.5–11 GHz direct modulation link using long-wavelength VCSEL, single-mode fiber, and high-speed photodetector
 - **Alternative approaches**
 - 0.5–11 GHz external modulation link
 - One fiber per link
 - One fiber per 2 links (WDM)
 - Downconversion with image rejection
 - Optical distribution of LO and IF signals
 - **Schedule**

Baseline Approach

Low-cost 0.5 – 11 GHz Direct Modulation Link



Issues:

- Best 11 GHz VCSEL performance projections currently offered by Cielo, Honeywell, Novalux, Sandia, Bandwidth9 → sufficiently low RIN, but what is input IP₂, IP₃?
- VCSEL performance is T-dependent → how to address?
- 11 GHz detector cost (\$1,300) dominant factor in total projected link cost of ~\$2,090
- Reliability (MTBF?, gain and phase stability?, withstand temperature?)



Directly Modulated Laser Options

	Part Number	Notes (price per 1000)
Alcatel	1915LMI	insufficient BW (2.5 Gb/s)
Avalon Photonics Ltd	L3125	insufficient BW (3.125 Gb/s)
Bandwidth9	VCSEL	currently insufficient BW (2.5 Gb/s), anticipate 10 Gb/s
Cielo Communications	10 Gb/s VCSEL	contacted for quote
Corning/Lasertron	QLM715-03	insufficient BW (3 GHz)
Fermionics	LCDIL	insufficient BW (< 1 GHz)
Fujitsu	FLD5F6CX-E	insufficient BW (2.5 Gb/s)
Gore	VCSEL	insufficient BW (1.25 Gb/s)
Honeywell	?	promise >10 GHz, but can't find
Infineon	V1310	12 Gb/s 1310 module; requested quote
Mitel	1A448	insufficient BW (2 GHz)
Mitsubishi	FJT1300	insufficient BW (< 5 GHz)
MODE (Emcore)	8685-1400 VCSEL array	insufficient BW (2.5 Gb/s or maybe less)
Nova Crystals	VCELs	insufficient BW (2.5 Gb/s)
Novalex	NECELS	insufficient BW (2.5 Gb/s) 10Gb/s out 7-2001
OKI	OL3301 or 3302	insufficient BW (< 1 GHz)
Ortel (Lucent)	Daytona D1861A	10 Gb/s 1310 module; requested quote
	1541B	13 GHz transmitter module; \$10,200 for 1
OptoSpeed	LCSH1550A	not fiber pigtailed
Sandia/Photonic Systems	N/A	not a product til 2002, > 10 GHz not anticipated

We anticipate < \$1,000 now , < \$750 by 2002

Lowest-cost solutions

We anticipate < \$1,000 now , < \$750 by 2002

~\$4,000, \$3,000 by 2002

> \$ 8,000 for 1000, \$ 7,500 by 2002

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Fiber Options

- **Single-Mode**
 - available for ~ \$0.15/meter (plus connector cost)
 - low loss (~0.3 dB/km) at either 1300 or 1550 nm
 - near-zero material and waveguide dispersion at 1300 nm (standard fiber) or 1550 nm (dispersion-shifted fiber)
 - low polarization mode dispersion
 - zero spatial mode dispersion
- **Multimode**
 - available for lower cost than Single-Mode
 - low loss (~0.3 dB/km) at either 1300 or 1550 nm
 - even for best graded-index MM fiber, spatial mode dispersion limits $BW \cdot L$ to < 2 GHz km, unless a mode stripper is used (which could offset any MM vs SM cost savings)

$BW \cdot L \gg 11$ GHz km

Photodetector Options

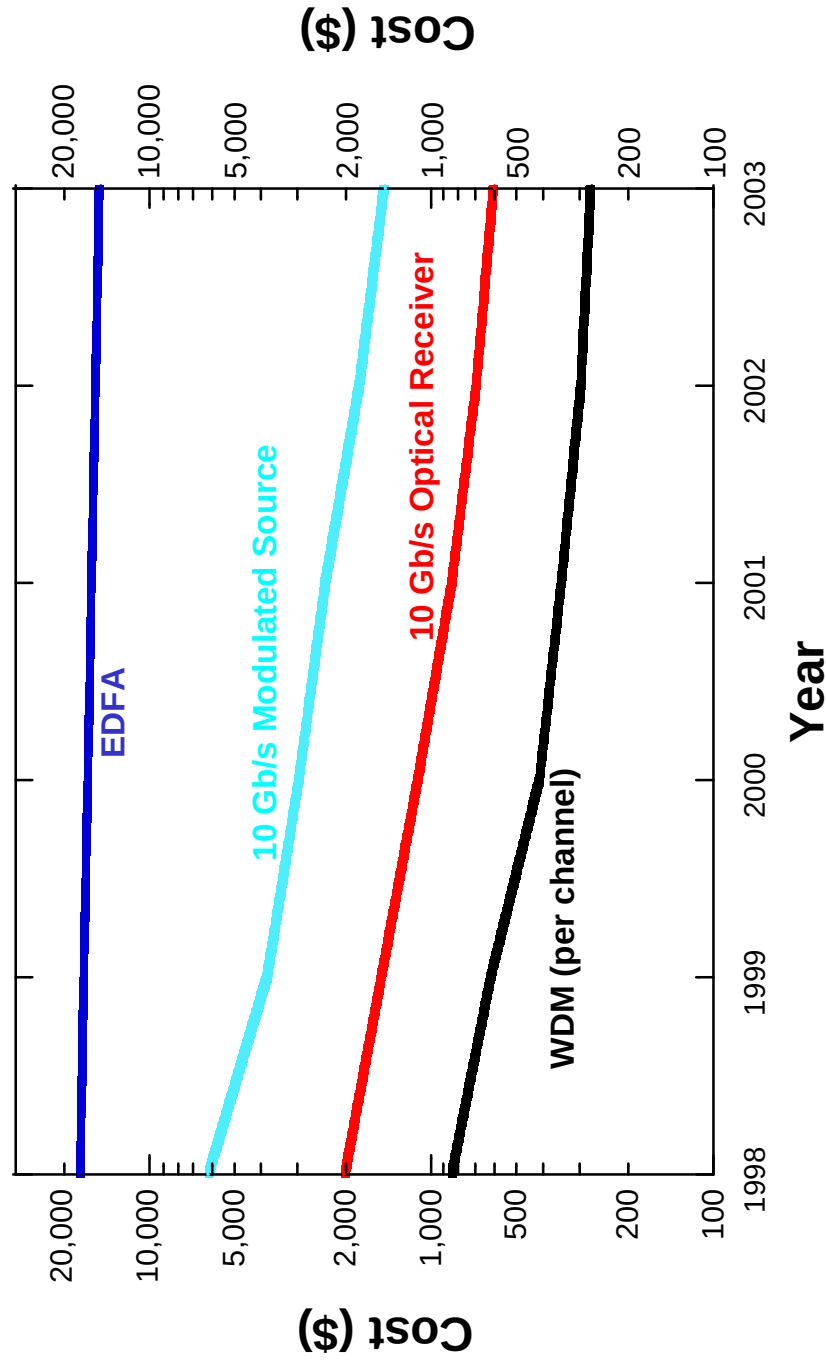
	Part Number	Notes (price per 1000)	
Alcatel	1900 DMC	insufficient BW (2.5 Gb/s)	
Corning/Lasertron	QDMH1-053	\$1,800	→ \$1,350 by 2002
Discovery	DSC-R402	\$1,760	→ \$1,340 by 2002
Fermionics	FD80S7-F	insufficient BW (5 GHz)	
Fujitsu	FRM5N142DS	\$3,900	→ \$2,975 by 2002
Multiplex	MTRX192A	waiting for quote	
NewFocus	PD1544	\$4,165	→ \$3,121 by 2002
Nortel	PP-10G	\$1,782	→ \$1,345 by 2002
Ortel (Lucent)	2518A	\$2,940 for 1, awaiting 1000 quote	
OptoSpeed	PDMH12A	requested quote	
PD-LD	PD	insufficient BW (2.5 Gb/s)	

Lowest cost photodetector is Discovery = \$1,340 per link by 2002

+ ~ \$ 750 for laser

= ~ \$2,090 for link

Forecast for Cost of Analog F/O Link Components*



Source: "Technology Roadmap on Microwave Photonics,"
Optoelectronics Industry Development Association, August 2000

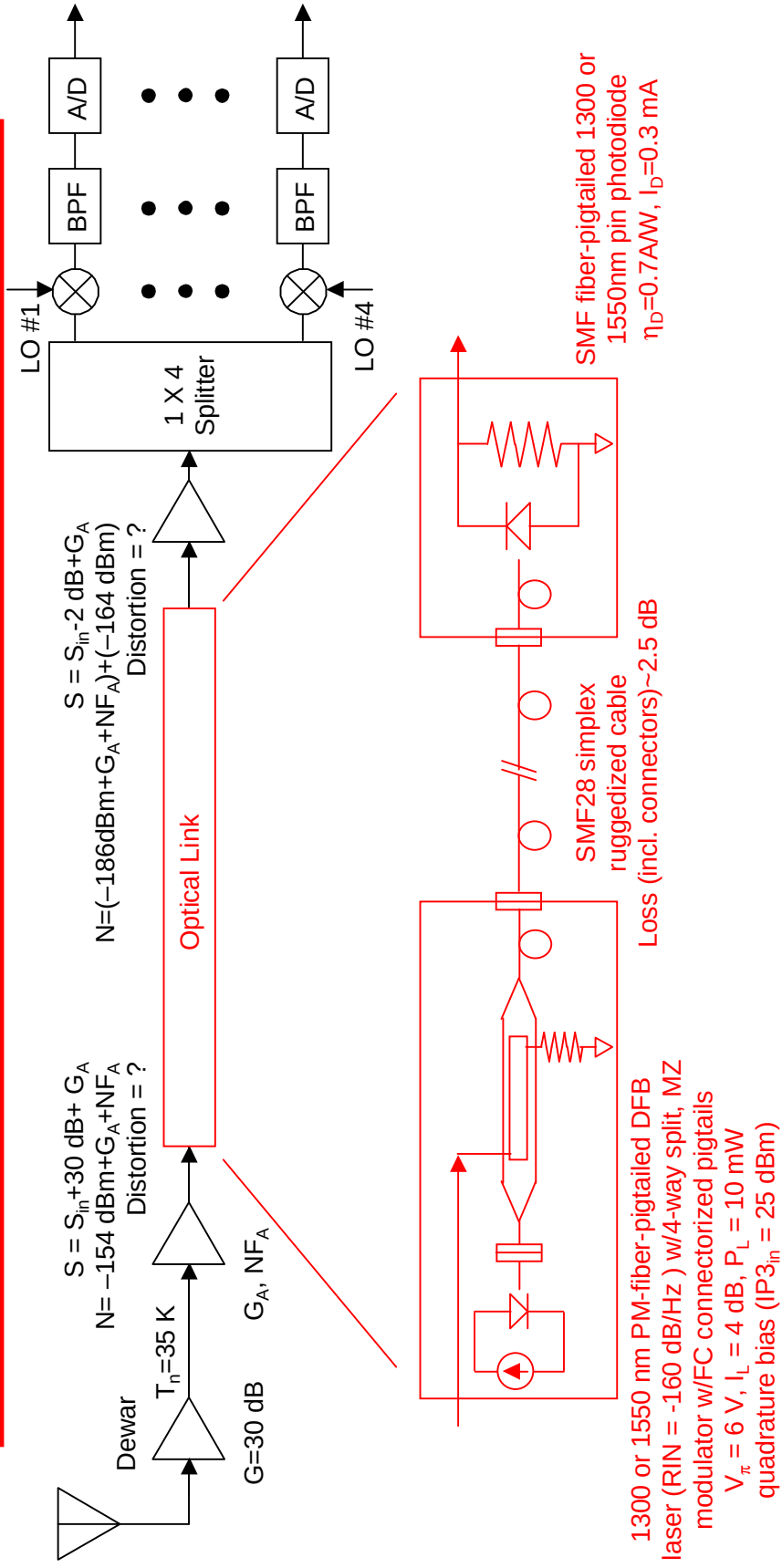
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Alternative Approach

External Modulation Link



Issues: • Total cost > \$3,000 per link (projected for 2002)



External Modulation Laser/Modulator Options

	Part Number	Price each for 1000	
Alcatel	1915LMM	integrated laser-EA; requested quote	
EO-Space	?	no product yet	
Fujitsu	FLD5F10NP	integrated laser-EA; \$4,000	~\$3,000 by 2002
GEC Marconi (Caswell)	LC1576	requested quote	
JDS Uniphase	10150-002171		\$2,500
Ericsson	PGT20405/01	integrated laser-EA; \$4,000	~\$1,875 by 2002*
Lasertron	?	10GHz integrated laser-EA next year	~\$3,000 by 2002
Lumera	?	product due out next year (polymer)	We anticipate ~\$2,000 by 2002*
Micro Photonix Integration	OC-192 mod		~\$2,250 by 2002*
Mitsubishi	ML9XX14	requested quote	
Multiplex	MTX 510EW	integrated laser-EA; requested quote	
NewFocus	Bulk EO	insufficient BW (<1 GHz)	
Nortel	TX1550	integrated laser-EA; \$5,981	~\$4,495 by 2002
Pacific Wave	?	product overdue (polymer)	We anticipate ~\$2,000 by 2002*
SDL	10AP-MOD9201	requested quote	
Srico	10 GHz mod		\$2,500
TriLink	OC-192 mod	10.7 GHz; \$2,800	~\$2,100 by 2002*

* requires a PM-fiber-pigtailed CW laser; Nortel sells one, which by 2002, split between 2 modulators, will cost about \$1,300 per link

Therefore, lowest-cost laser/modulator solution is Nortel laser + Lumera or Pacific Wave modulator = ~\$2,300 per link by 2002

+ ~\$1,340 detector
= ~\$3,740 for link

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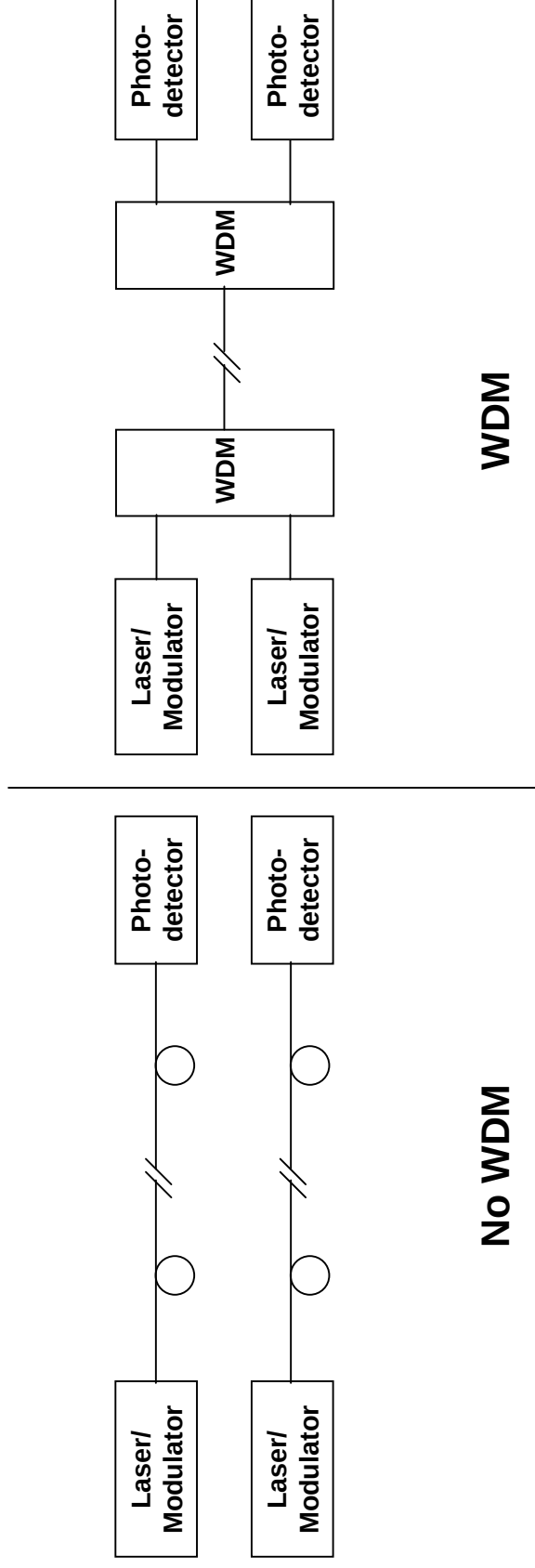
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Alternative Approach

External Modulation Link with Dense WDM

Cost of fiber is reduced if we use WDM to allow both polarizations of an antenna's signal to propagate in a single fiber.



Issues: Cannot be implemented with directly modulated lasers due to chirp

MUX/DEMUX cost prohibitive (> \$500/channel even in 2002)

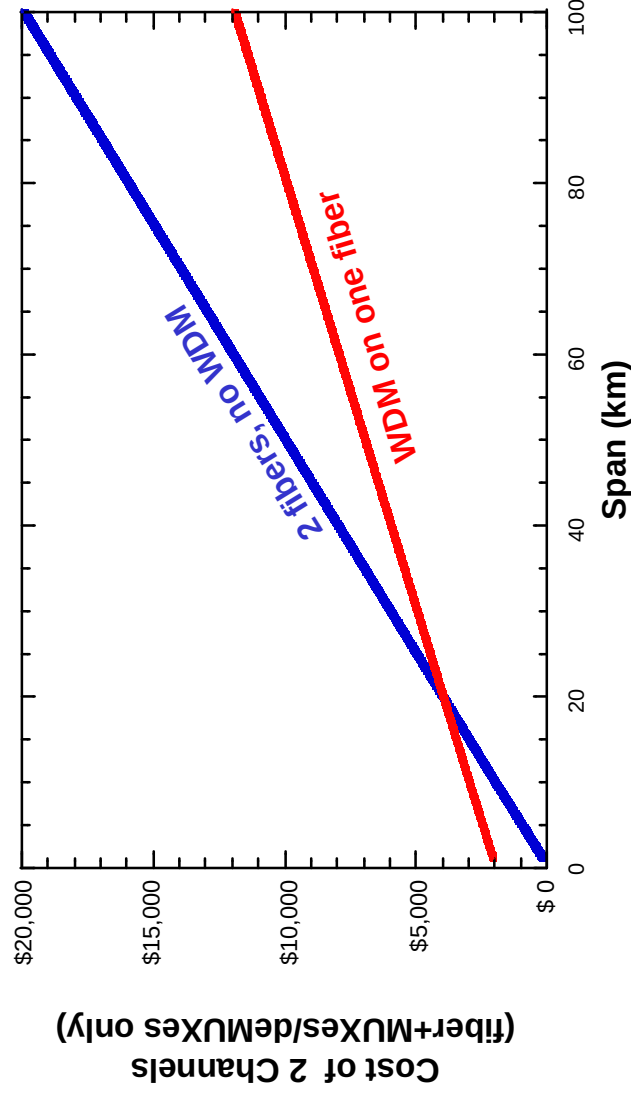
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Wavelength Multiplexing Considerations

If external modulation were to become cost-feasible, WDM could be used. However, significant cost savings come only at longer distances, given the low price/meter of fiber and the high price/channel of the WDMs.

Example: fiber cost \$0.20/meter; WDM cost \$1,000 (500/channel)



For > 20 km spans, 2-channel WDM yields a cost savings

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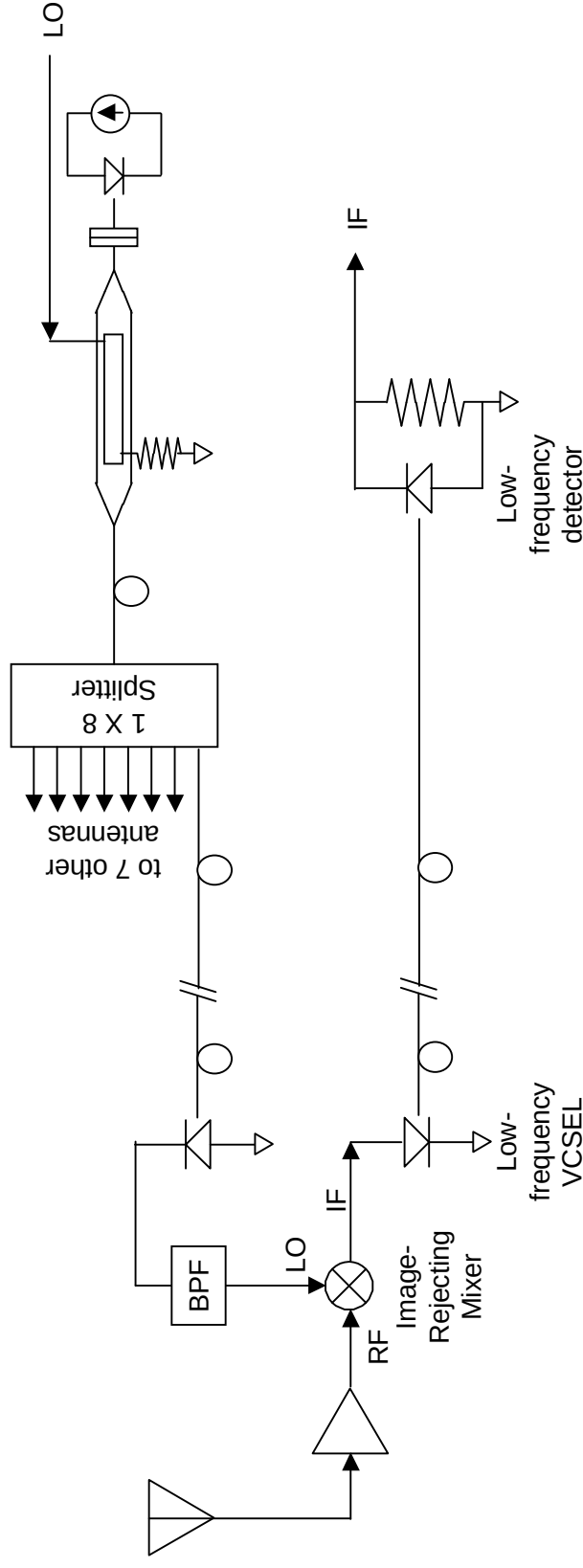
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Alternative Approach

Optical Downconversion with Image Rejection



Issues: Doesn't bring entire band to receivers simultaneously

Requires mixer performance improvement

Still costlier than baseline approach (~\$2,400 compared to baseline \$2,090)

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Schedule

- **Task 1: Study to identify design solution** (May 2000—December 2000)
 - 1.1 *Link definition*—Define antenna and receiver interfaces to the link, quantify goals.
 - 1.2 *Concept exploration*—Perform high-level evaluation of direct and external modulation broadband link designs, as well as SSB down-conversion link designs. Down-select to baseline approach.
 - 1.3 *Detailed architecture evaluations*—Perform detailed performance and cost analyses of baseline and backup link design approaches.
- **Task 2: Link development** (January 2001—April 2002)
 - 2-1 *Prototype demonstration*—Assemble and perform benchtop demonstration of all link components in the selected architecture. Provide all experimental data to SETI.
 - 2-2 *Package design*—Prepare conceptual drawings of the component packages.
 - 2-3 *Production plan development*—For each key component in the link, solicit proposals and quotes from candidate vendors, to show that we can meet the cost goal.
 - 2-4 *CDR*—Report on prototype design and performance at April 2000 CDR.
- **Task 3: Link fabrication for the PTA** (May 2002—December 2002)
 - 3-1 *Preproduction*—Layout and fabricate PC boards as required.
 - 3-2 *Module fabrication, testing, and installation*—Fabricate modules for PTA using PC boards and drawings from 3-1. Assist SETI personnel with installation on site.